



**THE UNIVERSITY
OF QUEENSLAND**
AUSTRALIA

This exam paper must not be removed from the venue

Venue _____

Seat Number _____

Student Number

Family Name _____

First Name _____

School of Mathematics & Physics
Semester One Examinations, 2022
MATH1061 Discrete Mathematics

This paper is for St Lucia Campus students.

Examination Duration: 120 minutes

Planning Time: 10 minutes

Exam Conditions:

- This is a Closed Book examination - no written materials permitted
- Casio FX82 series or UQ approved or labelled calculator only
- During Planning Time - Students are encouraged to review and plan responses to the exam questions
- This examination paper will be released to the Library

Materials Permitted in the Exam Venue:

(No electronic aids are permitted e.g. laptops, phones)

None

Materials to be supplied to Students:

Additional exam materials (e.g. answer booklets, rough paper) will be provided upon request.

None

Instructions to Students:

If you believe there is missing or incorrect information impacting your ability to answer any question, please state this when writing your answer.

Answer all questions in the space provided.

Credit can only be given for work written in this examination script.

SHOW ALL WORKING.

Solutions given without justification will not receive full marks.

Marks will be allocated for clear and logical arguments.

Use the back of pages if the space provided is insufficient or for rough working.

The number of marks per question is shown on the examination paper.

For Examiner Use Only

Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	

Total _____

1. The logical binary operation NAND, denoted $|$, is defined by

$$p | q \equiv \sim (p \wedge q).$$

- (a) **(2 marks)** Write down the truth table for $\sim (p | q)$. Show all working.
- (b) **(3 marks)** Prove, using truth tables or otherwise, that $(p | p) \equiv (\sim p)$. Show all working.
- (c) **(3 marks)** Prove, using truth tables or otherwise, that $((p | p) | (q | q)) \equiv p \vee q$. Show all working.

2. (a) Consider the statement

There exists a real number x such that for all real numbers y , $xy = 1$.

- (i) **(2 marks)** Write the negation of this statement.

- (ii) **(2 marks)** State which of the original statement or its negation is true. Justify your answer, showing all working.

- (b) **(4 marks)** Consider the statement

For all integers a and b , if $(ab)^2$ is odd then a is odd and b is odd.

Use either a direct proof, a proof by contraposition or a proof by contradiction to prove this statement is true. Show all working.

3. (a) **(2 marks)** Let $A = \{0, 1, 2, 3, 4\}$, $B = \{3, 4, 5, 6\}$ and $C = \{1, 2, 4, 5, 7\}$ and let

$$R = ((A - B) \cap (A - C)) \cup (B \cap C \cap A).$$

List the elements of the set R .

- (b) **(2 marks)** Let $X = \mathcal{P}(\emptyset)$ and $Y = \mathcal{P}(\mathcal{P}(\emptyset))$. List the elements of each of the following sets.

(i) X

(ii) Y

(iii) $X \cap Y$

(iv) $X \cup Y$

4. **(5 marks)** Prove that $|\mathbb{Z}^+ \times \mathbb{Z}^+| = |\mathbb{Z}^+|$.

5. (a) **(4 marks)** Let a be an integer such that the remainder is 1 when a is divided by 4. Prove that when a^4 is divided by 16 the remainder is also 1.
- (b) **(4 marks)** Use Euclid's Algorithm to verify that $\gcd(2000, 1713) = 1$. Clearly show all steps in the execution of the algorithm.

6. **(8 marks)** Use mathematical induction to prove that the following statement is true.

For all integers $n \geq 0$, $3^n < (n + 3)!$

7. Consider the function $f : \mathbb{Q} \cap [1, \infty) \rightarrow \mathbb{Q} \cap [2, \infty)$ given by

$$f(x) = (x - 1)^2 + 2 \text{ for all } x \in \mathbb{Q} \cap [1, \infty).$$

(a) **(4 marks)** Prove that the function f is injective.

(b) **(3 marks)** Prove that the function f is not surjective.

8. Consider the relation ρ on $\mathbb{Z}$ given by

$$a\rho b \text{ if and only if } (a-b)(a+b) = 4k \text{ for some } k \in \mathbb{Z}.$$

(a) **(6 marks)** Prove that ρ is an equivalence relation.

(b) **(2 marks)** Determine the number of equivalence classes of ρ . Justify your answer.

9. (a) **(1 marks)** Determine whether the groups $(\mathbb{Z}, +)$ and $(\mathbb{R} - \{0\}, \cdot)$ are isomorphic. Justify your answer.
- (b) **(3 marks)** Determine whether the groups $(\mathbb{Z}_4, +)$, $(\mathbb{Z}_5 - \{[0]\}, \cdot)$ are isomorphic. Justify your answer.
- (c) **(4 marks)** Find a subset S of $\mathbb{Z}_{10}$ such that $(S, \cdot)$ is a group of order 4 (where $\cdot$ denotes multiplication modulo 10). Justify your answer.

10. For this question, you may leave your answers as expressions involving factorials.

(a) **(2 marks)** How many distinct arrangements can be made of the letters of the word WOOLLOONGABBA?

(b) **(3 marks)** If the letters of the word WOOLLOONGABBA are permuted at random, what is the probability that we get the same word back?

(c) **(3 marks)** How many distinct arrangements are there of the letters of the word WOOLLOONGABBA such that exactly one of the following is true?

- The two *A*s are not next to each other.
- The two *B*s are not next to each other.
- The two *L*s are not next to each other.

11. (a) **(4 marks)** Let T be a tree on p vertices where $p \geq 11$, T has precisely six vertices of degree 1 and precisely four vertices of degree 3. Determine the degree of each of the remaining $p - 10$ vertices of T .
- (b) **(4 marks)** Let d be a positive integer and let T be a tree such that T has a vertex of degree d . Prove that T has at least d vertices of degree 1.

Working space next page.

Working space only.

END OF EXAMINATION